# CivicFix

Smart Solutions for Better Cities

Report **It**. Track **It**. Fix **It** .



# THE PROBLEM

Reports about potholes, damaged footpaths, broken streetlights and overflowing bins can be duplicated or lost across channels. Develop a platform to classify reports, merge likely duplicates, assign them to responsible teams and show status updates to residents.



POTHOLES & DAMAGED ROADS



BROKEN FOOTPATHS

# SOLUTION

Create one central platform where residents can submit geotagged reports, municipal teams can manage work orders, and everyone can monitor progress through a transparent status-tracking system

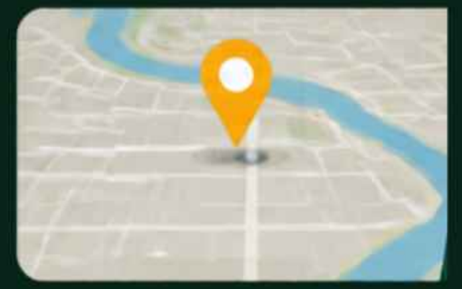


NON-WORKING STREETLIGHTS



OVERFLOWING PUBLIC BINS

# ONE REPORT.
# FIVE CLEAR STEPS.



**1 REPORT**
Residents report with photo and location




**2 CLASSIFY**
Our system identifies the issue type

ISSUE TYPE

POTHOLE

Road Infrastructure

PRIORITY

Medium

LOCATION

MG Road, Bengaluru

**3 DEDUPE**
We find likely duplicates to avoid repeat work

LIKELY DUPLICATES

MG Road
Reported 1 day ago
120m

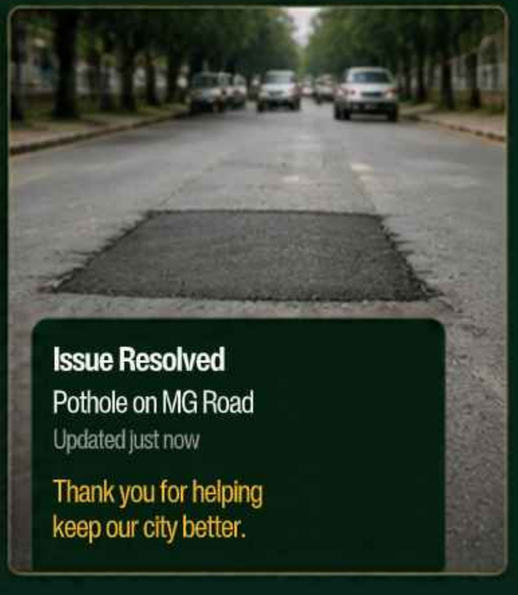
MG Road
Reported 2 days ago
180m

MG Road
Reported 3 days ago
250m

**4 ASSIGN**
The right team gets the right issue

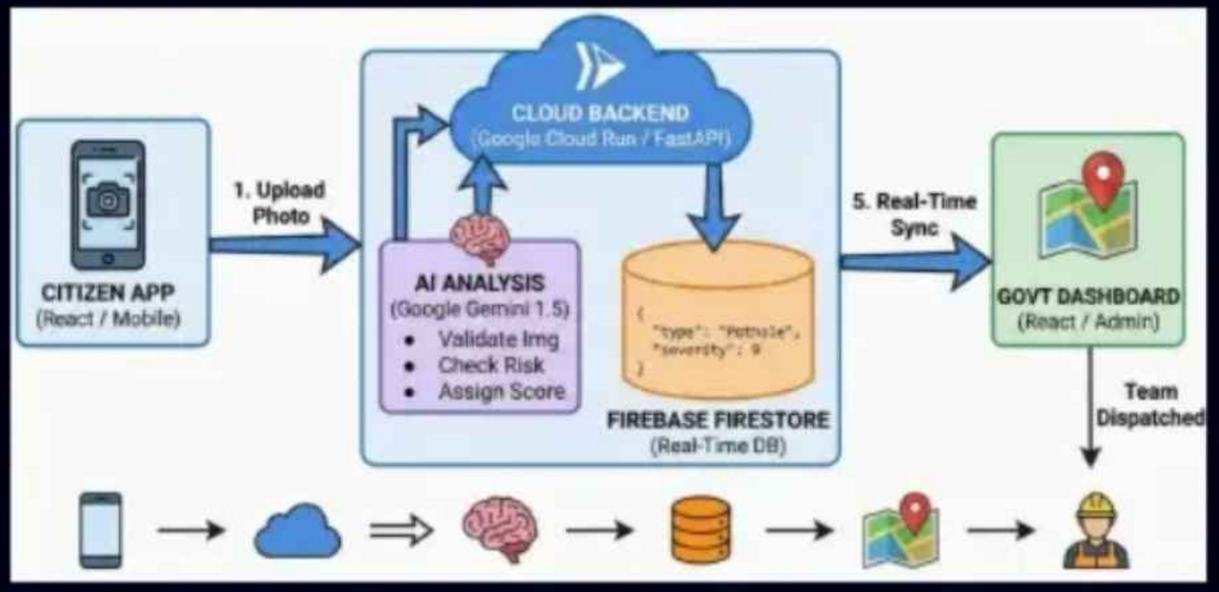



**5 RESOLVE**
Fix issues and update residents transparently



# HOW CIVICFIX WORKS

Complete System Workflow




1 **Citizen Upload**

Photo captured via React/Mobile app and uploaded to Cloud Backend

2 **AI Processing**

Gemini validates image, checks risk factors, assigns severity score

3 **Data Storage**

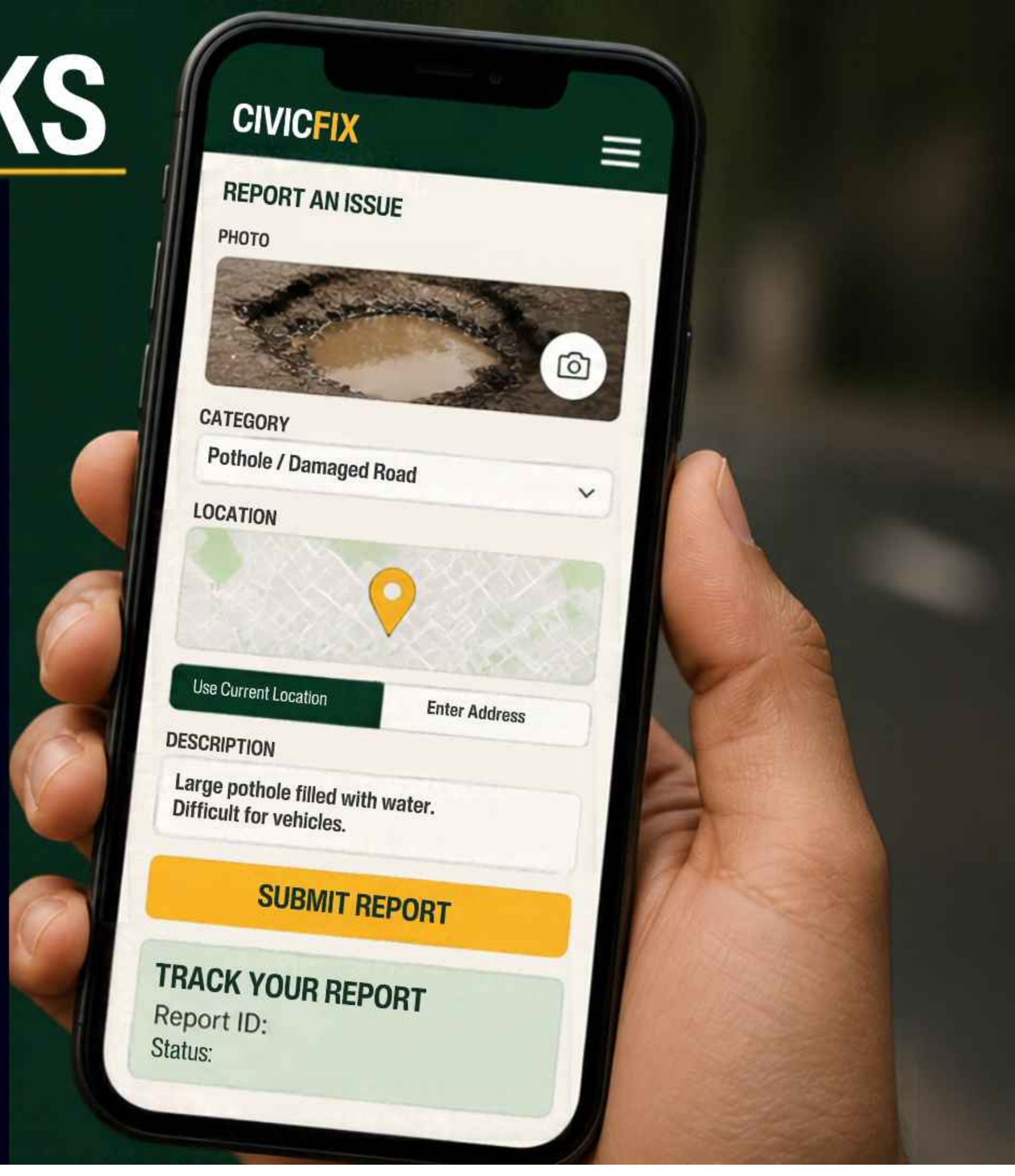
Structured JSON stored in Firestore with type, severity, location metadata

4 **Dashboard Sync**

Real-time update pushed to Government Admin Dashboard

5 **Team Dispatch**

Municipal workers receive prioritised ticket and location for immediate action



# Technology Stack

Unified platform for reporting and tracking

AI-powered issue classification

Location intelligence and mapping

Secure data storage and real-time alerts





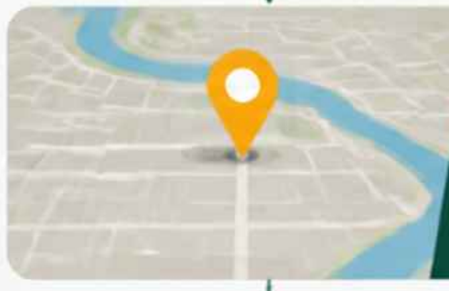

## Web & Mobile Interface

Report, track and get updates




## javaScript

Smart categorisation and routing



## Maps & Geolocation

Accurate location and visual insights



## Database & Notifications

Secure storage and real-time alerts

# DASHBOARD DESIGN

- Complaints by category and locality.
- Daily and weekly complaint trends.
- Average time from reporting to resolution.
- Duplicate reports merged.
- Overdue complaints by responsible department.
- Repeat infrastructure issues after repair.



# SMART TRIAGE, NOT MORE NOISE

We connect the dots, surface likely duplicates, and **let people make the final call.**

| | ORIGINAL REPORT | NEARBY REPORT 1 | NEARBY REPORT 2 |
|---|---|---|---|
| |  |  |  |
| DESCRIPTION | Big pothole with water | Large pothole, waterlogged | Deep pothole collecting water |
| CATEGORY | Pothole | Pothole | Pothole |
| LOCATION | MG Road, Bengaluru | MG Road, Bengaluru | MG Road, Bengaluru |
| TIME REPORTED | Today, 08:15 AM | Today, 08:42 AM | Today, 09:10 AM |
| SIMILARITY | 100% | 94% | 91% |

**HUMAN REVIEW**

High similarity detected
Review and confirm action

**MERGE THESE REPORTS**

**LOCATION DISTANCE**

**35 m**

Between reports

# A PRACTICAL, SCALABLE TECH STACK

Built for reliability today. Ready to integrate tomorrow.

**INTERFACES**

React | Citizen App | React | Staff Dashboard

**APPLICATION LAYER**

nodes | Express | REST API

**DATA LAYER**

MongoDB | Issues & Reports | Status & Updates

**INFRASTRUCTURE**

Maps & Geospatial | Image Storage | Notifications & Alerts

**EXTERNAL INTEGRATIONS**

- javaScript (issue classification, priority prediction)
- Third-party Services (weather, traffic, open data)
- Email, SMS & Messaging Gateways

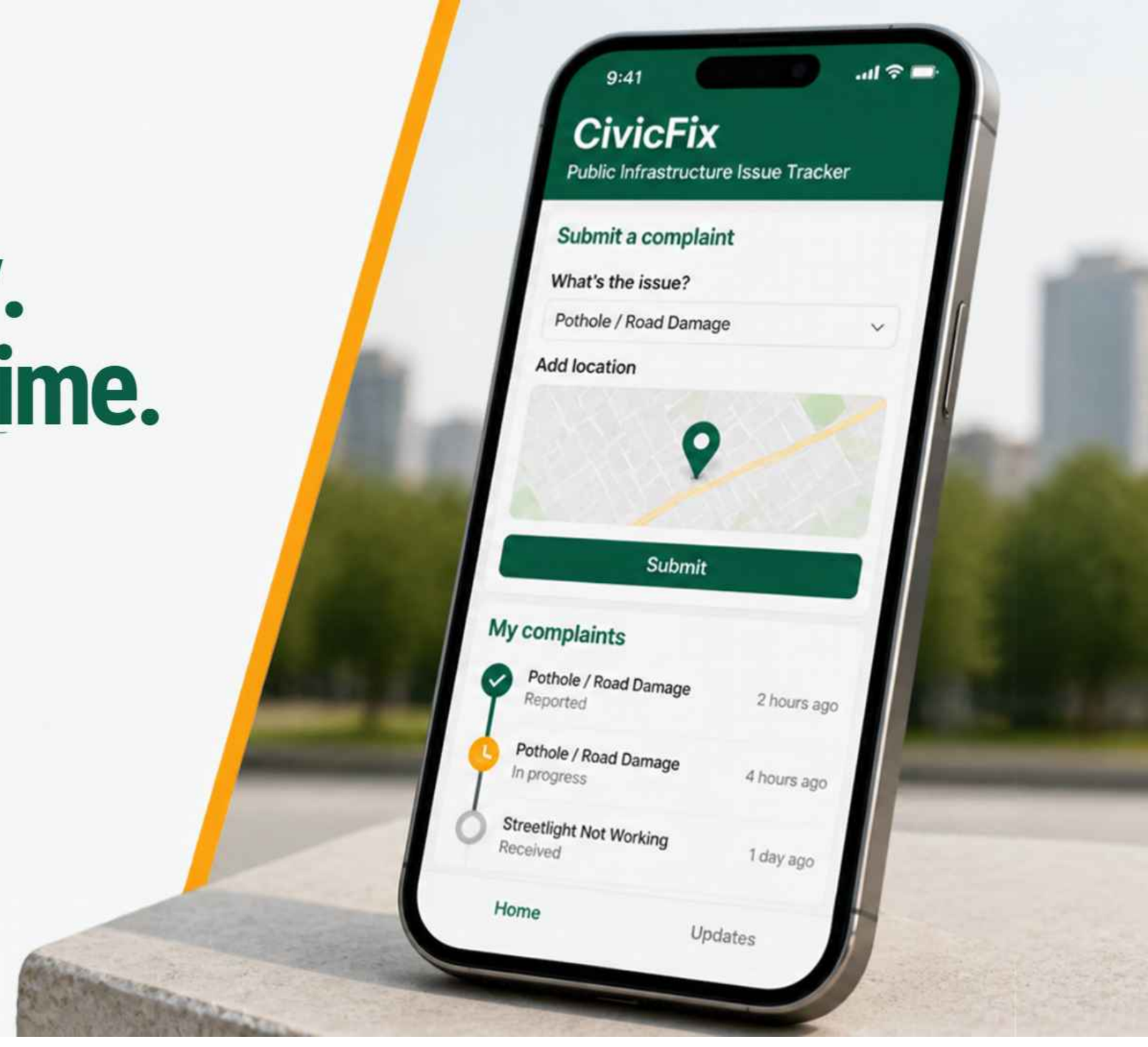
CITIZEN INTERFACE

# Report simply. Track in real time.

**Easy report submission**

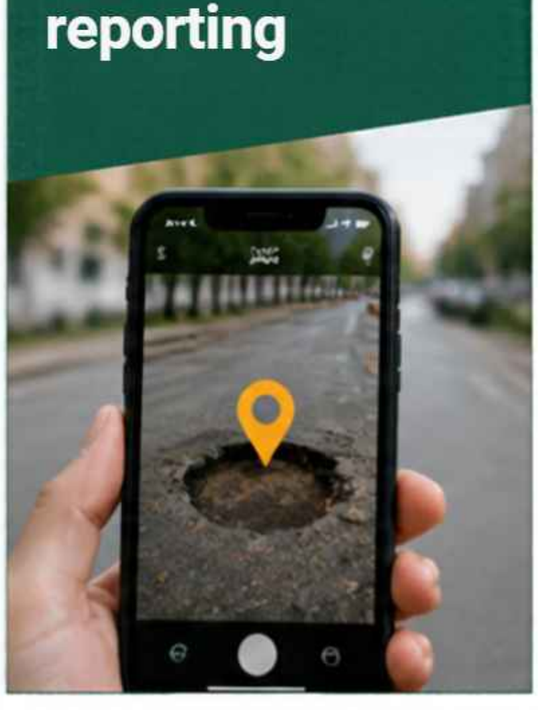
**Select issue category and drop a pin**

**Track submitted complaints**

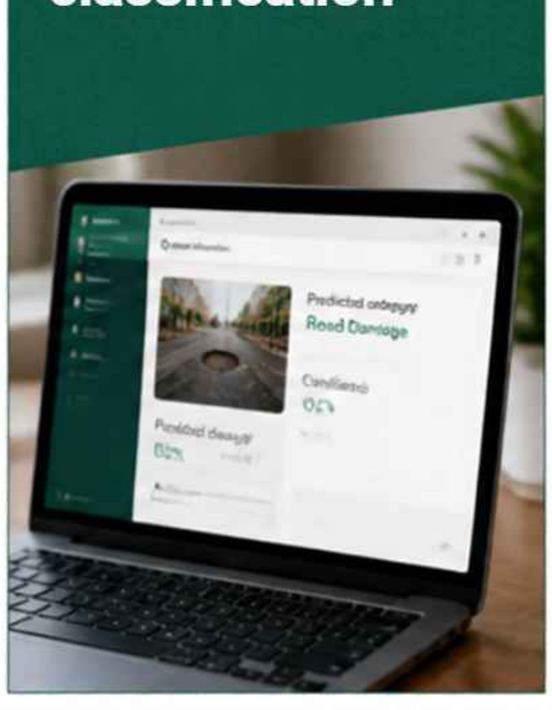


# Key Features.

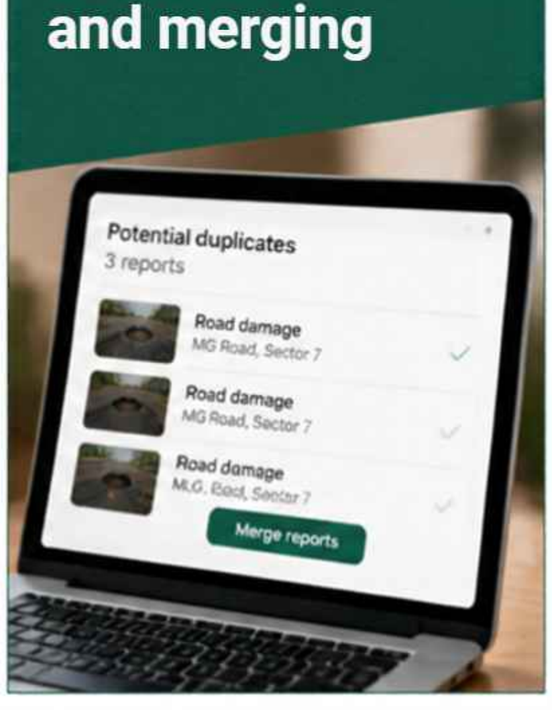
## Photo and location reporting



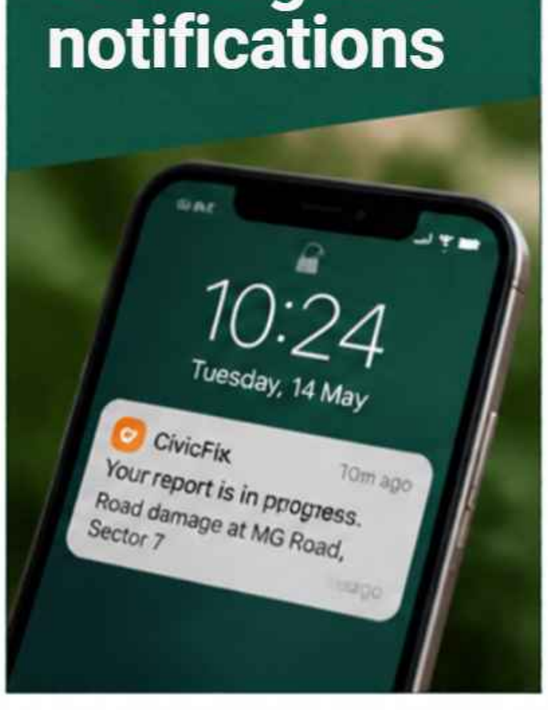
## AI-assisted classification




## Duplicate detection and merging




## Status tracking and notifications



# Impact & Benefits.

- Fewer duplicate reports
- Better team coordination
- More transparent governance
- Faster issue resolution



# Report It.
# Track It. Fix It .
# Build Better Cities.

**CivicFix – Public Infrastructure Issue Tracker**

**Smarter reporting. Clearer updates.**

**Stronger cities for everyone.**

**CivicFix** | Aligned with Ministry of Housing and Urban Affairs

